

Seunghyeon Seo

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RESEARCH INTERESTS

My research focuses on efficient and robust deep learning for real-world applications. I am particularly interested in **data-efficient 3D vision**, including NeRF and Gaussian Splatting under sparse inputs, with an emphasis on regularization, ray augmentation, and ray parameterization. I am also increasingly interested in **synthetic data generation and training with generative models** to improve robustness and generalization, while remaining broadly open to impactful problems across CV, ML, and related areas.

WORK EXPERIENCE

- Blizzard Entertainment, Irvine, CA | *Applied Research Scientist*** Feb. 2026 ~ Present
- Develop machine learning-driven solutions for 3D asset processing, improving efficiency and scalability of internal game development pipelines.
- Kakao Mobility, Seongnam, Korea | *Research Scientist*** Sep. 2025 ~ Feb. 2026
- Worked with multimodal perception data (e.g., camera images, LiDAR point clouds) for multimodal lane detection model based on BEVFusion for HD map generation.
 - Improved the efficiency of HD map annotation and manual correction workflows, reducing human-in-the-loop effort for lane-level map updates.
- Meta Reality Labs, Burlingame, CA | *Research Scientist Intern*** May 2025 ~ Aug. 2025
- XRCIA, Datasets (Mentors: John Kim, Lei Xiao, Beibei Liu)
 - Research about synthetic egocentric data generation with high-quality body extremities using video diffusion transformer models.
 - Leveraged Wan 2.1 video diffusion model with LoRA fine-tuning to adapt generative outputs for egocentric perspectives.
- Meta Reality Labs, Burlingame, CA | *Research Scientist Intern*** Jul. 2024 ~ Jan. 2025
- XRCIA, Datasets (Mentors: John Kim, Shaojie Bai, Tianyang Ma)
 - Research about synthetic data generation using conditional multi-view diffusion models, and training framework of universal face encoder leveraging real+synthetic data.
 - Successfully reduced the cost of data collection by over an order of magnitude compared to traditional real-world capture workflows.
 - Built a high-quality synthetic dataset used alongside real data to train a universal face encoder, resulting in over 5% improvement on key metrics, surpassing the current best model.
- ThinkforBL Consulting Group, Seoul, Korea | *Machine Learning Engineer*** Jun. 2020 ~ Nov. 2020
- Developed and applied deep learning-based perception and time-series models (e.g., Darkflow, CNN-LSTM) for livestock behavior analysis, crop growth pattern classification, and large-scale movement visualization.
 - Built end-to-end ML-driven services, implementing an item-based collaborative filtering recommender system and developing client-facing REST APIs using the Django REST Framework to support customization needs.

EDUCATION

- Seoul National University, Seoul, Korea** Mar. 2021 ~ Aug. 2025
- Ph.D. in Artificial Intelligence
- Seoul National University, Seoul, Korea** Mar. 2014 ~ Feb. 2021
- B.A. in Agricultural Economics / Data Sciences

AWARDS AND SCHOLARSHIPS

- Best Doctoral Dissertation Award Honorable Mention** Aug. 2025
- Seoul National University
- Outstanding Reviewer Award** Sep. 2024
- ECCV 2024
- Qualcomm Innovation Fellowship Korea 2023 Winner** Nov. 2023
- Qualcomm AI Research
- Youlchon AI Star Scholarship** Aug. 2023
- Youlchon Foundation & AI Institute-Seoul National University
- AI Fellowship (Fully Funded)** Mar. 2022 ~ Feb. 2023
- Seoul National University

PUBLICATIONS

Conferences / Journals

- [1] Ingyun Lee, Jae Won Jang, **Seunghyeon Seo**, Nojun Kwak, “[DivCon-NeRF: Diverse and Consistent Ray Augmentation for Few-shot NeRF](#)”, *IJCAI 2026*.
- [2] Shaojie Bai*, **Seunghyeon Seo***, Yida Wang, Chenghui Li, Owen Wang, Te-Li Wang, Tianyang Ma, Jason Saragih, Shih-En Wei, Nojun Kwak, Hyung Jun Kim, “[Generative Head-Mounted Camera Captures for Photorealistic Avatars](#)”, *SIGGRAPH Asia 2025. (Journal Track) [Project Page]*
- [3] **Seunghyeon Seo**, Yeonjin Chang, Jayeon Yoo, Seungwoo Lee, Hojun Lee, Nojun Kwak, “[ARC-NeRF: Area Ray Casting for Broader Unseen View Coverage in Few-shot Object Rendering](#)”, *CVPRW 2025. (Oral) [Project Page]*
- [4] Donghoon Han*, **Seunghyeon Seo***, Eunhwan Park, SeongUk Nam, Nojun Kwak, “[Unleash the Potential of CLIP for Video Highlight Detection](#)”, * indicates equal contribution, *CVPRW 2024. [Code]*
- [5] Yeonjin Chang, Yearim Kim, **Seunghyeon Seo**, Jung Yi, Nojun Kwak, “[Fast Sun-aligned Outdoor Scene Relighting based on TensoRF](#)”, *WACV 2024*.
- [6] Donghoon Han, **Seunghyeon Seo**, DongHyeon Jeon, Jiho Jang, Chaerin Kong, Nojun Kwak, “[ConcatPlexer: Additional Dim1 Batching for Faster ViTs](#)”, *NeurIPSW 2023. (Oral)*
- [7] **Seunghyeon Seo**, Yeonjin Chang, Nojun Kwak, “[FlipNeRF: Flipped Reflection Rays for Few-shot Novel View Synthesis](#)”, *ICCV 2023. [Project Page] [Code]*
- [8] **Seunghyeon Seo**, Jaeyoung Yoo, Jihye Hwang, Nojun Kwak, “[MDPose: Real-Time Multi-Person Pose Estimation via Mixture Density Model](#)”, *UAI 2023. [Code]*
- [9] Jaeyoung Yoo*, Hojun Lee*, **Seunghyeon Seo**, Inseop Chung, Nojun Kwak, “[End-to-End Multi-Object Detection with a Regularized Mixture Model](#)”, * indicates equal contribution, *ICML 2023. [Code]*
- [10] **Seunghyeon Seo**, Donghoon Han*, Yeonjin Chang*, Nojun Kwak, “[MixNeRF: Modeling a Ray with Mixture Density for Novel View Synthesis from Sparse Inputs](#)”, * indicates equal contribution, *CVPR 2023. (Qualcomm Innovation Fellowship Korea 2023 Winner) [Project Page] [Code]*
- [11] Jongmook Kim, Jooyoung Jang, **Seunghyeon Seo**, Jisoo Jeong, Jongkeun Na, Nojun Kwak, “[MUM: Mix Image Tiles and UnMix Feature Tiles for Semi-Supervised Object Detection](#)”, *CVPR 2022. [Code]*
- [12] Kyuewang Lee*, Inseop Chung*, Daeho Um, Jaeseok Choi, Yeji Song, **Seunghyeon Seo**, Nojun Kwak, Jin Young Choi, “Multi-modal Object Detection, Tracking, and Action Classification for Unmanned Outdoor Surveillance Robots”, *ICCAS 2021*.

Preprints

- [1] Donghoon Han, Eunhwan Park, **Seunghyeon Seo**, “[Visual Words Meet BM25: Sparse Auto-Encoder Visual Word Scoring for Image Retrieval](#)”, Under Review. [\[Project Page\]](#)
- [2] Yeonjin Chang, Juhwan Cho, **Seunghyeon Seo**, Wonsik Shin, Nojun Kwak, “[LoGoColor: Local-Global 3D Colorization for 360 Scenes](#)”, Under Review. [\[Project Page\]](#)
- [3] Yeonjin Chang, Erqun Dong, **Seunghyeon Seo**, Nojun Kwak, Kwang Moo Yi, “[ROODI: Reconstructing Occluded Objects with Denoising Inpainters](#)”, Under Review. [\[Project Page\]](#)

PROJECTS

Research on Novel View Synthesis Using NeRF Trained with Sparse Viewpoint Data

Funded by Samsung Electronics | Main Researcher

Jul. 2023 ~ Jul. 2024

- Neural rendering, NeRF, Few-shot learning
- Conducted research to improve the performance of NeRF under sparse input conditions, resulting in a published research paper at a top-tier computer vision conference.

Artificial Intelligence Research about Cross-Modal Dialogue Modeling for One-on-One Multi-Modal Interactions

Funded by Ministry of Science and ICT of Korea | Research Assistant

May 2022 ~ Jun. 2023

- Multi-modal learning, Object detection
- Assisted in building an object detection system that processes user-submitted images to identify clothing items and extract key attributes.

Development of Real-Time Multi-Camera Object Tracking and Identification Technology

Funded by Electronics and Telecommunications Research Institute | Main Researcher

Jun. 2021 ~ Dec. 2021

- Multi-object tracking
- Developed a multi-view, multi-object tracking algorithm for real-time vehicle and pedestrian tracking within a parking lot environment.

Development of Multimodal Sensor-Based Intelligent Systems for Outdoor Surveillance Robots

Funded by Ministry of Science and ICT of Korea | Research Assistant

Jan. 2021 ~ Aug. 2021

- Multi-modal learning, Object detection
- Contributed to a research project on multi-modal object detection, tracking, and action classification for autonomous outdoor surveillance robots.

PATENT

Method and Apparatus based on NeRF using Flipped Reflected Ray, KR102808144B1

SKILLS

Programming Languages

- Python, R, CUDA

ML Development Stack

- PyTorch, Jax, Tensorflow, Slurm, MAST Scheduler, Docker

Language

- English (Professional), Korean (Native), French (Basic)

TALK

Generative Head-Mounted Camera Captures for Photorealistic Avatars Jun. 2025

- Meta Reality Labs

Novel View Synthesis from Sparse Inputs via NeRF Apr. 2025

- SNU Haedong Advanced Engineering

MixNeRF: Modeling a Ray with Mixture Density for Novel View Synthesis from Sparse Inputs Jul. 2023

- SNU Research Achievement Symposium

ACADEMIC SERVICE

Reviewer for TCSVT, AAAI '25/'26, BMVC '26, CVPR '23/'24/'25/'26, ECCV '24/'26, ICCV '25, NeurIPS '25/'26, SIGGRAPH Asia '26